

Article

Hydrogen-Rich Syngas Production from Waste Textile Gasification Coupling with Catalytic Reforming under Steam Atmosphere

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Abstract: The average annual global production of waste textiles exceeds 92 million tons, with the majority landfilled and incinerated, resulting in energy waste and environmental pollution. In this study, a thermal conversion process for waste textiles by gasification coupling with catalytic reforming under a steam atmosphere was proposed. The gasification performance of the waste textiles jumped with the introduction of steam and catalyst compared to pyrolysis at 800 °C. The syngas yield increased from 20.86 to 80.97 mmol/g and the hydrogen concentration increased from 17.79 to 50.91 vol.%, which was an increase of 288.12% and 186.18%, respectively. The excellent gasification performance mainly came from two sources: steam promotion for volatiles production and Fe-N-BC promotion for steam reforming of volatiles by Fe₂O₃, Fe₃O₄, Fe-N_x, etc. This study has achieved the efficient production of hydrogen-rich syngas from waste textiles, providing a new idea and theoretical basis for the effective removal and utilization of waste textiles.



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Keywords: waste textiles; gasification; catalytic reforming; steam atmosphere; hydrogen-rich syngas

1. Introduction

Over the past two decades, the production and average annual consumption of textile have doubled in response to population growth, economic development, and the rise of fast fashion [1]. More than 92 million tons of waste textile are generated globally each year, mainly landfilled or incinerated [2], with only about 15% being recycled [3], posing a huge environmental burden and resource challenge to modern society [4]. Due to the customizable material properties of blends of natural and synthetic fibers [5], cotton and polyester have become the main fiber components of waste textiles [6], corresponding to the chemical composition of cellulose and polyethylene terephthalate (PET), respectively [7]. The complex composition is a key obstacle to the recycling of waste textiles. The recycling of waste textiles by conventional chemical means such as enzymatic digestion [8], ammonolysis [9], and glycolysis [10] first requires a troublesome separation process, and tends to cause secondary pollution and a large consumption of chemicals. Therefore, a thermal conversion process, which is simple and non-additive, has been considered for the effective removal and utilization of waste textiles.

Gasification, widely used for biomass, is an important way for the thermal conversion of complex organic wastes to generate high-value products [11,12]. Any carbon-based feedstock can be converted by gasification into a gaseous fuel, called syngas, at high temperatures and under controlled oxygen or steam conditions [13]. Syngas consists mainly of H₂ and CO and can be used directly as a fuel or industrial feedstock [14]. More research

attention is now being given to the gasification of waste textiles [15]. It is noteworthy that the co-gasification of plastics and biomass has a synergistic effect and can effectively change the composition of tar and syngas [16]. Similarly, the synergistic effect of cellulose and PET co-gasification was found by thermogravimetric and mass spectrometry (TG-MS) analysis [17]. Therefore, the gasification of waste textiles is a valuable thermal conversion process that utilizes the compositional properties of the feedstock itself.

Gasification under a steam atmosphere (steam gasification) has the advantage of high hydrogen yields and low by-product (char/tar) yields and is usually used for hydrogen-rich syngas production [18,19]. But little research has been conducted in this area, mainly focusing on the weight loss processes based on thermogravimetric curves [20,21]. There are few studies on the gasification process performance, and their results are not satisfactory. The performance of steam gasification on waste polyester-cotton was studied using a fluidized bed reactor at 850 °C [22]. The results showed that the hydrogen concentration (23.2 vol.%) of the syngas was unsatisfactory, despite the high LHV (10.17 MJ/Nm³). The hydrogen and syngas yield can be further improved by gasification coupling with catalytic reforming under a steam atmosphere (catalytic steam gasification) [23,24], but few studies have been conducted using this process to convert waste textiles. There are many types of catalysts for steam reforming [25], with active phases including noble metals [26], transition metals [27], alkali/alkaline earth metals [28], supports including Al₂O₃ [29], carbon [30], SiO₂ [31], and biochar-based [32] supports. Biochar-based catalysts have gained widespread attention due to their high yield, excellent thermal stability, rich surface functional groups, adequate pore structure, and environmental friendliness [33–35], and their catalytic performance can be further enhanced by metal loading, such as using Fe and Ni [36–38]. In addition, *Chlorella vulgaris*, with its high yield, is becoming an ideal natural material for the preparation of industrial biochar carriers due to its own high concentration of nitrogen that can form more active sites through self-doping [39]. Therefore, an excellent performance of waste textile gasification is expected to be obtained using Fe-loaded biochar-based catalysts prepared from *Chlorella vulgaris*.

In this study, a thermal conversion process for waste textiles by gasification coupling with catalytic reforming under a steam atmosphere was proposed. Firstly, the gasification performance of waste textiles was improved by introducing steam and further improved by parameter optimization. Then, Fe-N-BC with good catalytic activity was prepared for catalytic reforming, and the efficient production of hydrogen-rich syngas was achieved. Finally, the mechanism for waste textile gasification coupling with catalytic reforming under a steam atmosphere was clarified.

2. Materials and Methods

2.1. Materials

The waste polyester-cotton collected from Guangdong Province, China, was selected as the waste textile to be studied. The compositional analysis of materials is shown in Table S1. Before the experiment, the waste textiles were cut into small squares (5 × 5 mm), dried in an oven at 80 °C for 24 h, and then placed in a spare parts drying tray. The *Chlorella vulgaris* powder (≥98%) was purchased from Shanghai Guangyu Biotechnology Co., Ltd. (Shanghai, China). The iron chloride hexahydrate (FeCl₃·6H₂O) and trichloromethane were purchased from Aladdin Biochemical Technology Co., Ltd. (Shanghai, China).

2.2. Catalyst Preparation

Fe-N-BC was synthesized using a simple process (Figure 1) [40,41]. Typically, 5 g of FeCl₃·6H₂O, 15 g of *Chlorella vulgaris* powder (mass ratio 1:3), and 400 mL of deionized water were placed in a beaker, and then stirred with a magnetic stirrer for 4 h. The mixture was then dried and pyrolyzed in a horizontal tube furnace at 800 °C for 2 h under N₂ flow at a heating rate of 5 °C/min. Finally, the product was ground into powder and recorded as Fe-N-BC. Notably, all Fe-N-BC not additionally described in this study were prepared by this method. For comparison, pristine microalgae biochar (BC) and Fe-N-BC in different

mass ratios (1:1, 1:4, and 1:6) were synthesized using the same method. The catalysts used in the experiments had a particle size range of 90–150 μm .



Figure 1. Synthesis scheme of Fe-N-BC.

2.3. Experimental Setup and Procedure

The catalytic steam gasification experiment was conducted in a vertical fixed-bed reactor, which was divided into two parts, the upper and lower parts, which were the gasification and catalytic reforming zone, respectively (Figure 2). The usual experimental procedure was as follows. Firstly, 0.2 g of catalyst was placed in the catalytic reforming zone, where the secondary reforming reaction would take place. Then, 0.5 g of sample was added into the top of the quartz tube (outside of the furnace, where the temperature was below 80 °C), and the N₂ flow rate (40 mL/min) and temperature increase procedure (20 to 800 °C at 10 °C/min) were set. The peristaltic pump was turned on, and 0.1 g/min flow rate of distilled water was fed into the furnace to form steam flow after the set temperature reached. After 5 min for atmosphere stabilization, a piston was used to push the samples instantaneously into the gasification zone to achieve rapid gasification under a steam atmosphere, and the primary products were reformed in the catalytic zone by steam. The furnace and peristaltic pump were switched off after 30 min. The above procedure was complete catalytic steam gasification process, recorded as Experiment 1. Gasification experiments under steam and nitrogen atmosphere without catalyst were also carried out, recorded as Experiments 2 and 3. In order to achieve the actual catalytic effect, the effect of biochar carrier consumption needed to be eliminated. Nothing was added to the gasification zone, and all other conditions remained the same, recorded as Experiment 4. For comparison, syngas was collected using a gas bag and detected using gas chromatography (GC). To ensure the reliability of the data, each experiment was repeated three times, and the average was taken.

2.4. Characterization

The composition and concentration of syngas were determined by gas chromatography (GC9790II, Fuli Instruments, Wenling, China). The tar composition was analyzed by gas chromatography-mass spectrometry (GC-MS, Shimadzu QP2020, Duisburg, Germany). The Fe concentration of the fresh Fe-N-BC was calculated by inductively coupled plasma-optical emission spectrometry (ICP-OES, Agilent ICP730, Santa Clara, CA, USA). The acid properties of the fresh Fe-N-BC were assessed by temperature-programmed desorption of NH₃ (NH₃-TPD, Chem Bet 300, Quanta chrome Corporation, Boynton Beach, FL, USA). The analysis of fresh and used Fe-N-BC was performed as follows: The crystal composition was obtained using an X-ray diffraction analyzer (XRD, Empyrean Rheology, Parnco, The Netherlands). The microstructure was characterized by scanning electron microscopy (SEM, Hitachi Quanta650, Tokyo, Japan). The surface area and pore structure were determined by an auto sorption device (Autosorb-iQ, Quanta chrome Corporation, USA) using the N₂ adsorption/desorption isotherm at 77 K of catalyst. The chemical species were analyzed by X-ray photoelectron spectroscopy (XPS, Thermo Scientific K-Alpha+, Waltham, MA, USA) using an Al K α X-ray source for excitation as a reference of C1s peaks at binding energy 284.8 eV for calibration.

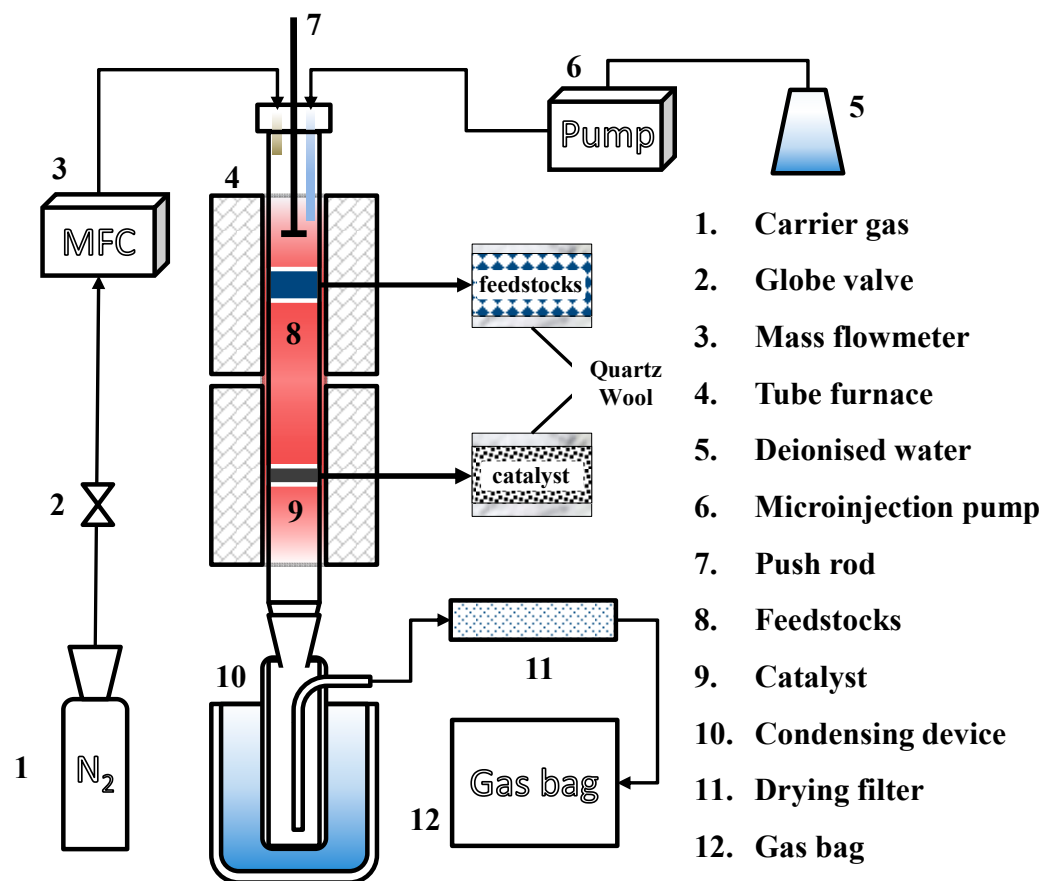


Figure 2. Schematic of catalytic steam gasification system.

2.5. Data Analysis

The gasification performance was characterized by gas concentration and syngas yield, and the tar conversion performance was characterized by the tar conversion rate [40]. The gas concentration (C_i , vol.%) was the volume percentage of the gas composition in syngas, and in this study, the hydrogen concentration was mainly used.

The gas concentration was calculated as following equation:

$$C_i = \frac{x_i}{x_{H_2} + x_{CO} + x_{CO_2} + x_{CH_4} + x_{C_2H_m}} \quad (1)$$

where x_i (vol.%) is the percentage of gas volume.

The syngas yield (Y_i , mmol/g) was calculated using the following equation:

$$Y_i = \frac{V_{N_2}}{22.4 \times m_{feedstock}} \times \frac{x_i}{x_{N_2}} \quad (2)$$

where V_{N_2} (mL) is the volume of N₂ introduced into the thermal transformation system, and $m_{feedstock}$ (g) is the mass of gasified sample.

The (T , %) was calculated using the following equation:

$$T = \frac{m_{tar}}{m_{blank}} \times 100\% \quad (3)$$

where m_{tar} (g) is the mass of tar produced by the experimental condition, and m_{blank} (g) is the mass of tar produced by the blank condition. Specifically, the blank group is the gasification of waste textile without steam (N₂ atmosphere) when the experimental variable is the steam flow rate, and the blank group is the gasification of waste textile without catalyst when the experimental variable is the type of catalyst.

3. Results and Discussion

3.1. Characterization of Fresh Catalyst

The Fe_2O_3 and Fe_3O_4 phases and more Lewis acidic sites were found in the Fe-N-BC compared to the BC (Figure 3a,b). It had been reported that both Fe_3O_4 and Fe_2O_3 have catalytic effects, mainly in the activation of H_2O to generate free radicals, promoting WGS and tar reforming reaction [42–44]. The Lewis acidic sites were extremely active to catalytic cracking of tar macromolecules, especially O-containing functional groups [45]. The microstructures of the catalysts were observed by SEM (Figure 3c,d), showing clear carbon microsphere structures in BC, which remained well intact morphology after loading Fe. This indicated that the Fe active particles were well dispersed in the char carrier and a good catalytic performance was expected [46]. An XPS analysis of the Fe-N-BC was carried out, and the elements Fe and N were focused on. The $2p^{3/2}$ - $2p^{1/2}$ spin-orbit bimodal peaks of oxidized Fe correspond to the Fe^{2+} and Fe^{3+} phases, respectively, which correspond to the iron oxide phase in the XRD results (Figure 3e). Figure 3f shows that the N 1s spectra were curve-fitted with five peaks with binding energies of 398.41, 398.80, 400.05, 401.04, and 402.71 eV, attributed to pyridinic N, Fe- N_x , pyrrolic N, graphitic N, and oxidized N, respectively [47]. This indicated that nitrogen atoms had entered the carbon network through in situ doping, and the different forms of nitrogen corresponded to the different doping sites in the carbon network. Since nitrogen (3.04) had a greater electronegativity than carbon (2.55), all kinds of N made the neighboring C atoms electron-deficient and reduced the gaps of the Fermi energy levels and conduction bands [48,49]. The carbon network was polarized by nitrogen doping, causing electron diffusion and the creation of electron holes, thereby increasing the adsorption capacity of biochar carrier [50]. Biochar itself had C-C, C-O, and C-H active sites, and a sparse structure that could serve as a good adsorbent for volatiles. It could be seen that C-N active sites with good adsorption capacity and charge migration efficiency strengthened the adsorption and reforming of volatiles. In addition, stable Fe- N_x active sites had been found, and N might change the σ -bond length between the radical and the active center by affecting the electron density of Fe, which resulted in a good catalytic performance of Fe- N_x [51,52]. It was worth noting that Fe compounds in the catalyst could increase the electron transfer between the H_2O and the Fe- N_x center, lowering the limiting energy barrier for radical formation and further improving the catalytic performance [41]. Based on the above analysis, Fe_2O_3 , Fe_3O_4 , and Fe- N_x were identified as the main active species from Fe loading, and the sparse structure, C-C, C-O, C-H, and C-N were the sources of catalytic performance of the char carriers.

3.2. Performance of Waste Textile Gasification

3.2.1. Performance of Gasification under Steam Atmosphere

The performance of waste textile gasification under a steam atmosphere was explored. Figure 4 shows the gasification performance of waste textiles at different steam flow rates and temperatures. The production of syngas, especially hydrogen, was promoted by steam in the atmosphere (Figure 4a). Both the syngas yield and hydrogen concentration were at their maximum levels at a steam flow rate of 0.1 g/min, 27.78 mmol/g, and 24.24 vol.%, respectively, which were increased by 33.34% and 36.24%, respectively, compared to the N_2 atmosphere. The decrease in performance (steam flow rate > 0.1 g/min) was due to the excessive steam lowering the reaction point temperature and decreasing the reaction time of volatiles [53–55]. The gasification performance increased with the rising temperature regardless of the atmosphere. The syngas yield and hydrogen concentration were improved by the steam at all temperatures, and the improvement increased with the rising temperature (Figure 4b). At the highest experimental temperature (900 °C), the syngas yield and hydrogen concentration increased to 40.22 mmol/g and 36.76 vol.%, respectively, which were 57.59% and 71.24% higher than in the N_2 atmosphere. The results indicated that the steam reforming reaction of volatiles was facilitated by a high temperature, resulting in the generation of gases, especially hydrogen.

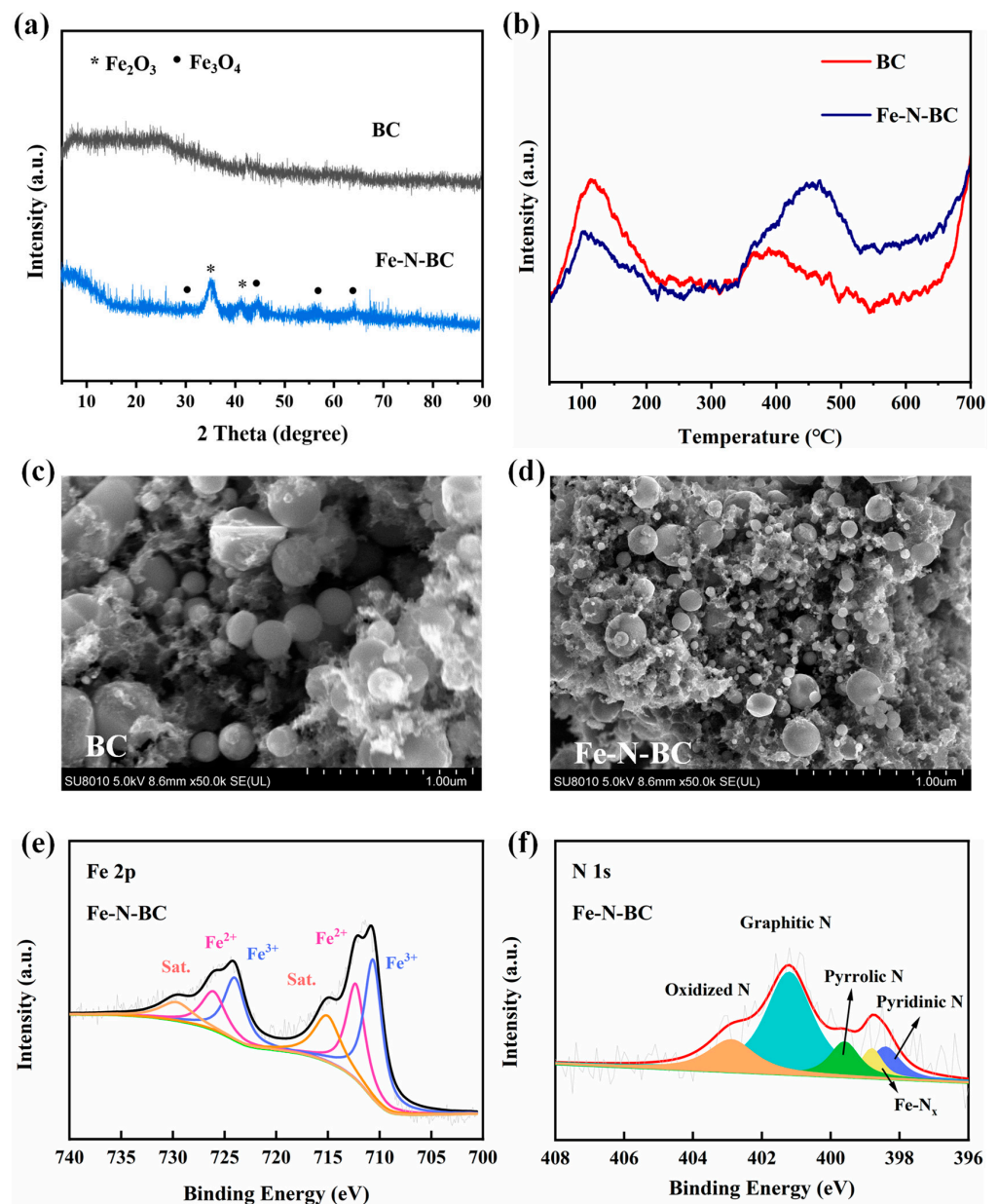


Figure 3. Characterization of fresh catalyst: XRD (a), NH_3 -TPD (b), SEM (c,d), and XPS (e,f).

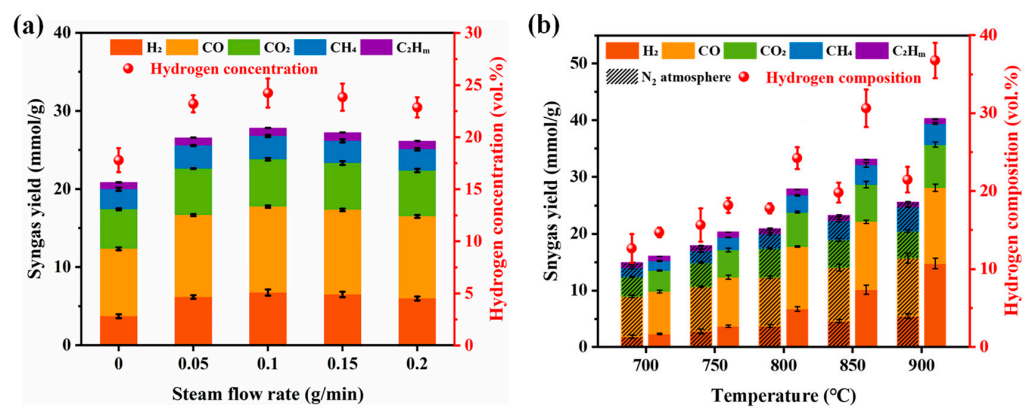


Figure 4. Performance of gasification under steam atmosphere at different operating conditions: (a) steam flow rate (800°C); (b) temperature (0.1 g/min).

3.2.2. Performance of Catalytic Gasification under Steam Atmosphere

Biochar-based catalysts were introduced to reduce energy consumption and produce hydrogen-rich syngas. It was obvious that the syngas yield and hydrogen concentration were increased by the BC and significantly increased by the Fe-N-BC (Figure S1). The catalyst and operating parameters of the Fe-N-BC catalytic steam gasification system were optimized. The Fe loading was first explored, and the catalyst amount, steam flow rate, and temperature were initially set at 40 wt.%, 0.1 g/min, and 800 °C, respectively, according to Section 3.1 and previous research [56,57]. The best catalytic performance was achieved at a mass ratio of $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ to microalgae of 1:3, which meant that the Fe loading was 15.20 wt.% according to ICP-OES (Figure 5a). Using the above conditions as initial values, further optimization was carried out for catalyst amount, steam flow rate, and temperature. Figure 5b–d show that the best catalytic steam gasification performance was achieved when the catalyst amount, steam flow rate, and temperature were 60 wt.%, 0.1 g/min, and 900 °C, respectively. It is worth noting that due to the partial gasification of the biochar carrier during the reaction, the syngas from the carrier consumption (obtained by Experiment 4) had to be deducted when optimizing the catalyst amount. The best performance was achieved at a catalyst amount of 40 instead of 60 wt.% (Figure S2). This was because excessive catalysts led to more gas production from biochar carrier gasification, inhibiting gas production from the feedstock. In addition, when the temperature reached 800 °C, the syngas yield and hydrogen concentration were improved to 80.97 and 50.91 vol.%, respectively. The increase in gasification performance slowed down with further temperature increase, especially the hydrogen concentration, which was almost unchanged (Figure 5d). And both the carbon conversion efficiency and gasification efficiency exceeded 100% (Table S2). The above findings indicate that the promotion of gasification was reduced and there was significant catalyst consumption after 800 °C [40]. Therefore, 800 °C was selected as the optimal temperature considering the lower energy and catalyst consumption. In conclusion, the best catalytic steam gasification performance was achieved when the Fe loading, catalyst amount, steam flow rate, and temperature were 15.20 wt.%, 40 wt.%, 0.1 g/min, and 800 °C, respectively, with the syngas yield and hydrogen concentration reaching 80.97 and 50.91 vol.%, respectively.

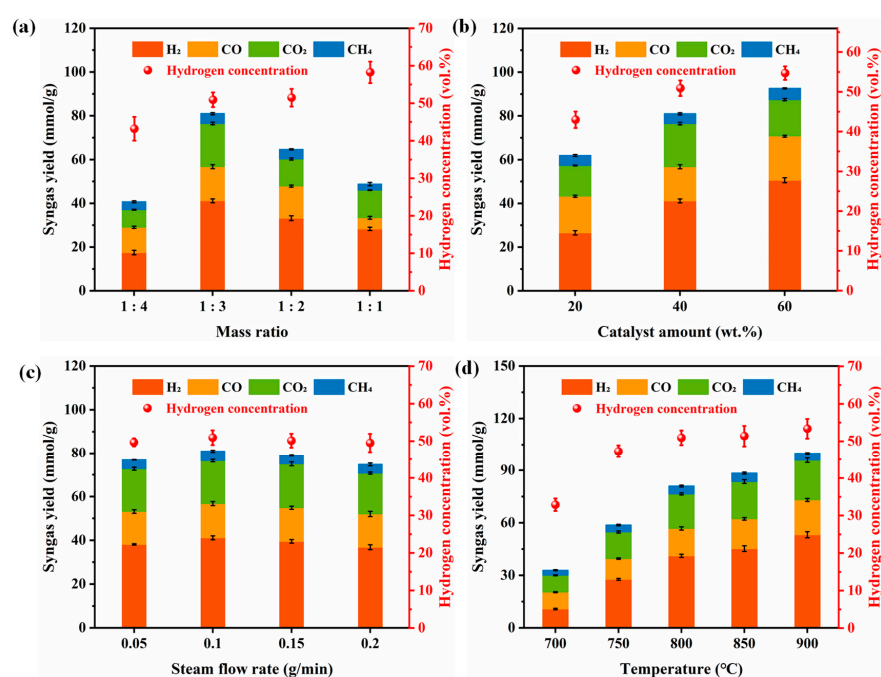


Figure 5. Performance of gasification coupling with catalytic reforming under steam atmosphere at different situations: (a) mass ratio ($\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$: microalgae); (b) catalyst amount; (c) steam flow rate; (d) temperature.

3.3. Mechanism

3.3.1. Mechanism of Gasification under Steam Atmosphere

Figure 6a shows the TG-DTG curves for different steam flow rates at 10 °C/min heating rate. Peak 1 and Peak 2 in the pyrolysis stage moved slightly towards lower with the increase in steam flow rate, and both peaks decreased slightly. These indicated that the pyrolysis temperature of the waste textiles was reduced, while the pyrolysis rate was suppressed with increasing steam. The lower pyrolysis rate might be caused by the steam inhibiting the dehydration reaction of some groups, such as hydroxyl groups prevalent in cellulose [58]. In contrast to the pyrolysis stage, the increase in Peak 3 (gas–solid reaction) at a higher steam flow rate indicated that the gasification reaction of residual char was promoted, which was due to the accelerated consumption of char (4) by steam reforming. According to the TG-DTG curves, higher material weight loss, meaning more volatile removal, was achieved with increasing steam.

Figure 6b shows the tar composition and conversion rates for different steam flow rates at 800 °C. Since the main fiber components of the waste textile were PET and cellulose (Table S1), the main components of the tar were polycyclic aromatic hydrocarbons (PAHs), derived from the benzene ring of PET and reforming of cellulose. The tar conversion rate increased and then decreased, reaching a maximum value of 60.06% at 0.1 g/min, with increasing steam. This was the same trend as that of the syngas yield (Figure 4a), indicating that the syngas yield was increased by the steam reforming of tar (5). From the tar composition, it can be seen that the increase in steam reduced the yield of components with O-containing functional groups, such as ketones, phenols, and acids, while generating H₂, CO, CO₂ [18,59], etc., which implied a lower oxygen content and higher LHV of the tar as a fuel. The increase in steam promoted the breaking of C–C bonds and C–O bonds, so that the yields of the tar, especially PAHs, decreased. Notably, the yield of 2-ring PAHs decreased and then increased with increasing steam. It was assumed that the thermal decomposition of excess steam generated a large number of H· radicals, which inhibited tar cracking [53,60]. Not only char and tar, but also syngas underwent steam reforming reactions such as the water–gas shift reaction (WGS, 6) and alkane reforming (7), which further increased the hydrogen concentration (Figure 4a). In summary, the steam mainly promoted the pyrolysis of waste textiles and the reforming of pyrolysis products (char, tar and gas) to produce more gases, especially hydrogen.

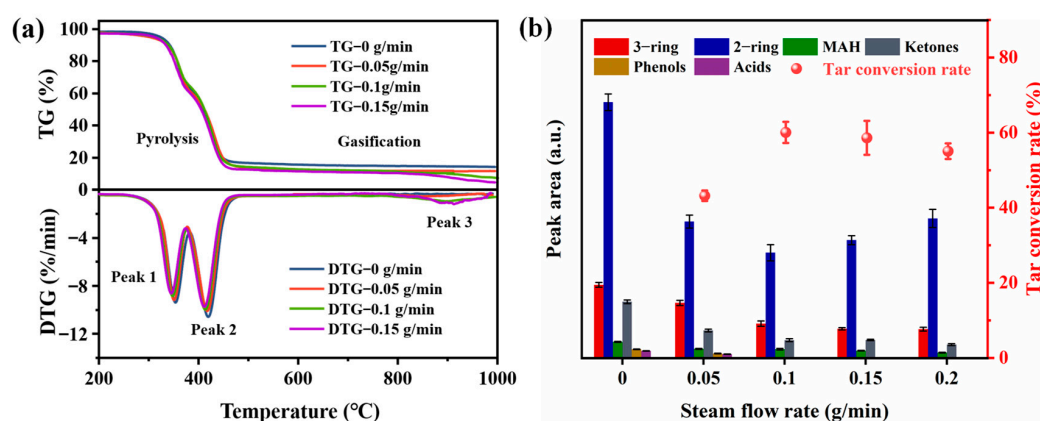
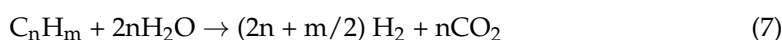


Figure 6. The TG–DTG curves (a) and tar (b) under different steam flow rates.

3.3.2. Mechanism of Gasification Coupling with Catalytic Reforming under Steam Atmosphere

(1) Characterization and performance of used Fe-N-BC

The used Fe-N-BC was characterized, and the changes in the Fe-N-BC after the reaction were clarified. The Fe-N-BC had a good stability, and the physical phase remained basically unchanged after the reaction (Figure S3a). The double peaks of the $2p^{3/2}$ – $2p^{1/2}$ spin orbitals of the oxidized state of Fe corresponded to Fe^{2+} and Fe^{3+} species, and the slight rightward shift in the peaks after the reaction represented the conversion of Fe^{2+} to Fe^{3+} (Figure S3b). It could be inferred that the catalyst was slightly oxidized after the reaction and part of Fe_3O_4 was converted to the Fe_2O_3 phase by steam oxidation. The possible mechanism for the catalytic tar reforming by iron oxides can be clarified as follows: (1) The tar adsorbed on the surface of iron oxide phase was oxidatively cleaved by lattice oxygen, leading to the creation of oxygen vacancies [61,62] and internal lattice oxygen migration due to concentration gradients [63], and the iron oxide phase was reduced. (2) The reduced phase was oxidized by the steam atmosphere, enabling lattice oxygen regeneration and H_2 production. Based on the SEM results, irregular broken particles could be clearly seen on the surface of the used Fe-N-BC (Figure 7a), which were considered to be deposited coke formed by hydrocarbon reforming [64] and might affect the mass transfer caused by clogging of the pore structure [65]. However, the change in the pore structure of the Fe-N-BC was not significant (Table S3), which was attributed to the fact that its stability was enhanced by the interaction of iron oxides with the biochar carrier [40], and the degradation of the biochar carrier by volatiles and steam was reduced. The catalytic performance of the used Fe-N-BC also remained stable due to the stability of the physical phase and pore structure. The catalytic performance of the used Fe-N-BC (syngas yield net of char losses) was only reduced by 8.23% and maintained by 85.16% at the third reaction (Figure 7b).

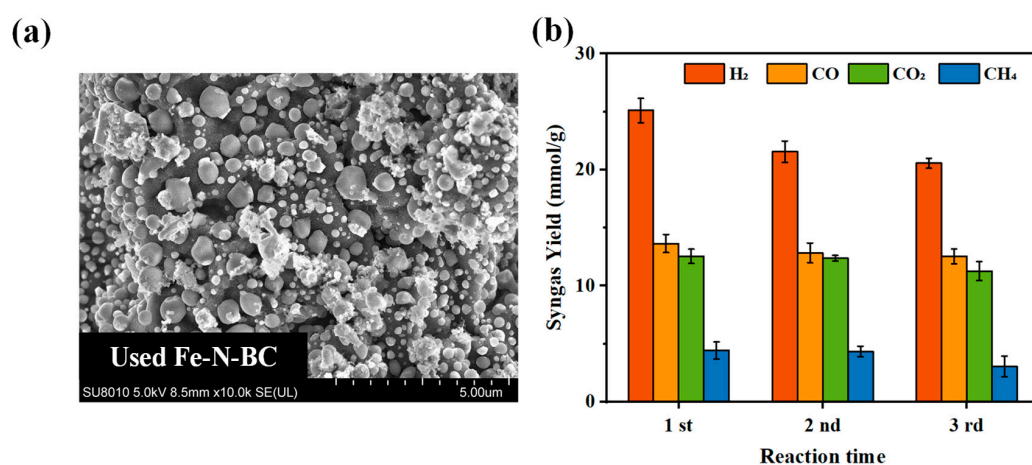


Figure 7. SEM after reaction (a) and performance stability (b) of Fe-N-BC.

(2) Tar production and actual catalytic effect

By comparing the gasification products using different catalysts, the catalytic mechanism could be further investigated and supports the previous conclusions. Figure 8a shows the tar production of different catalysts. The lowest tar conversion, with only 26.6%, was achieved using the BC. The catalytic activity was significantly improved after loaded with Fe, achieving a tar conversion of 84.3%. Furthermore, the addition of the biochar-based catalysts changed the tar composition. In the absence of a catalyst, the tar consisted mainly of aromatics (2-ring, 3-ring PAHs and MAH) and ketones. In the presence of the catalyst, the peak areas of the ketones, 2- and 3-ring PAHs decreased, and the peak area of the MAH slightly increased, implying the destruction of the O-containing functional groups and the cracking of the heavy tar. The above indicated that the BC itself promoted the tar conversion due to self-contained active sites, mainly C-C, C-O, C-H, and C-N [66–68], which was enhanced by Fe loading, corresponding to Fe_2O_3 , Fe_3O_4 , and $Fe-N_x$.

Biochar-based catalysts were partially consumed and converted to gas in catalytic reactions. In order to understand the catalytic effect, it was necessary to exclude the effects caused by catalyst gasification. Therefore, the actual catalytic effect of the catalyst was obtained by subtracting the gas yields of Experiment 2 and Experiment 4 from the catalytic gasification results (Figure 8b). The results show that BC promoted the production of H_2 , CO_2 , and CH_4 but inhibited the production of CO , indicating that BC promoted the 6 and selectively converted CO to H_2 . Similarly to the BC, the Fe-N-BC promoted the growth of H_2 and CO_2 significantly more than that of CO . The strongest catalytic effect on all gases was shown by the Fe-N-BC compared to the BC, corresponding to a high tar conversion rate (Figure 8a), attributed to Fe loading. In addition, C_2H_x disappeared in the biochar-based catalyst presence, indicating the promotion of light alkane reforming (7).

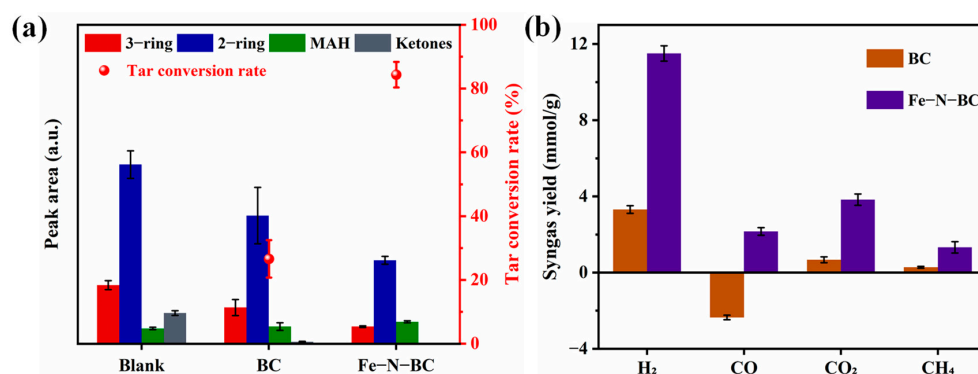


Figure 8. Tar production (a) and actual catalytic effect (b) using different catalyst.

(3) Mechanism of gasification coupling with catalytic reforming under steam atmosphere

Based on the previous analysis, it could be established that the excellent hydrogen-rich syngas production came from two sources. On the one hand, the pyrolysis of waste textiles and gasification of residual char were promoted by steam, leading to the generation of more volatiles, which were initially transformed by steam to light tar and H_2 . On the other hand, the steam reforming of volatiles was facilitated by the Fe-N-BC, mainly through the lattice oxygen transfer from Fe-oxides (Fe_2O_3 and Fe_3O_4) to tar, activation of steam by $Fe-N_x$, and catalysis of 5–7 reactions by C-C, C-O, C-H, and C-N active sites in the char carrier. In summary, the mechanism of catalytic steam gasification was mainly the steam promotion of volatile generation and steam reforming of volatiles by Fe_2O_3 , Fe_3O_4 , $Fe-N_x$, C-N, C-O, etc., as shown in Figure 9.

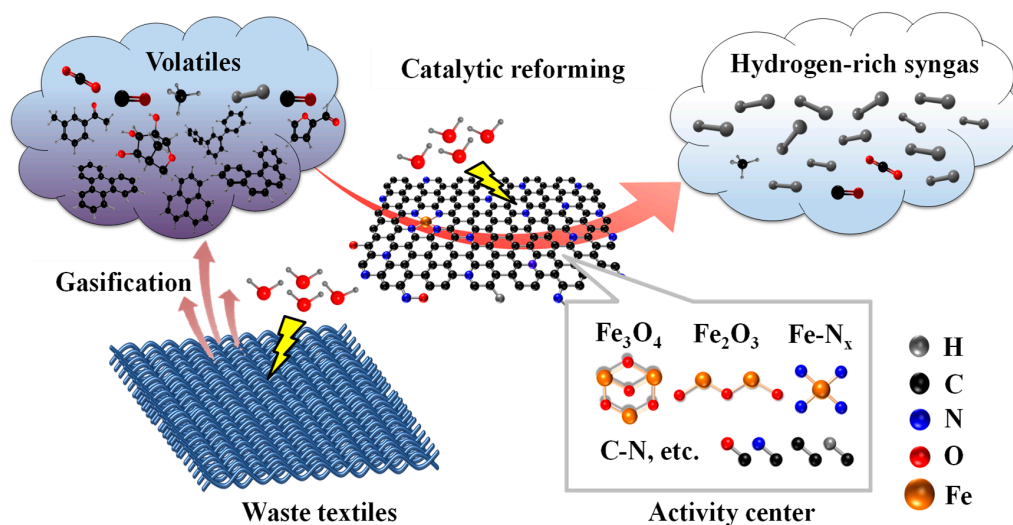


Figure 9. Mechanism of gasification coupling with catalytic reforming under steam atmosphere.

3.4. Discussion

Gasification is a promising direction for the thermal conversion of waste textiles, yet little research has been conducted so far. Currently, the main focus is on the study of gasification characteristics and kinetics under air or CO₂ atmosphere, with a poor syngas yield and high CO₂ yields [15,20,21]. Syngas quality can be effectively improved by steam, yet still much tar [69], and biochar-based catalysts can be used for tar elimination [70,71]. Therefore, a thermal conversion process for waste textiles by gasification coupled with catalytic reforming under a steam atmosphere was proposed. Gasification under a steam atmosphere without catalyst achieved a syngas yield of 27.78 mmol/g at 800 °C, which is similar to the result of others at 850 °C, and with a higher concentration of fuel gas (78.21 vs. 64.20 vol.%) [22]. The gasification performance was further improved by using Fe-N-BC, with syngas yield and hydrogen concentration of 80.97 mmol/g and 50.91 vol.% at 800 °C, which were higher than others' results of 39.5 mmol/g and 41.77 vol.% using CaO [72]. The above results indicate that the gasification coupled with catalytic reforming under a steam atmosphere process has a higher conversion efficiency and lower energy consumption, which is very promising to be extended for practical applications.

For the hydrogen-rich syngas obtained, the membranes [73,74] can be considered for fuel separation and purification to promote the subsequent efficient utilization in future research. Taking hydrogen with the highest value as an example, its application directions include applications in fuel cells [74–76], in combustion as fuel [77], in methanol preparation through Fischer–Tropsch synthesis [78,79], energy storage and grid balancing [80,81], and so on. In addition, scaling up the waste textile gasification system could be considered to facilitate the realization of practical applications, as well as exploring the nature of the materials for which the process is applicable and trying to extend the process to other solid waste.

4. Conclusions

In this study, a thermal conversion process for waste textiles by gasification coupling with catalytic reforming under a steam atmosphere was proposed. First, the gasification performance was enhanced after introducing steam into atmosphere, and the syngas yield and hydrogen concentration were 27.82 mmol/g and 24.24 vol.%, which were 33.34% and 36.24% higher than without steam. Then, the production of hydrogen-rich syngas was realized by the introduction of the Fe-N-BC, and the syngas yield and hydrogen concentration reached 80.97 mmol/g and 50.81 vol.%, respectively, which were 191.08% and 110.05% higher than without catalyst. Finally, the mechanism of gasification coupling with catalytic reforming under a steam atmosphere can be clarified as follows: (1) The generation and initial reforming of volatiles was promoted by steam. (2) The steam reforming of volatiles was promoted by the Fe-N-BC, mainly through the transformation of lattice oxygen from Fe-oxides (Fe₂O₃ and Fe₃O₄) to tar, the activation of H₂O by Fe-N_x, and the C-C, C-O, C-H, and C-N catalytic sites in the biochar. In this study, the efficient production of hydrogen-rich syngas from waste textiles was achieved through thermal conversion, and the main mechanism was clarified. This study provides a new idea and theoretical basis for the effective removal and utilization of waste textiles.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/pr12091790/s1>, Table S1. Ultimate, proximate and fiber analysis of textile waste sample ^a; Table S2. CCE and GE of catalytic steam gasification under different temperature [82–84]; Table S3. Structural characteristics analysis of Fe-N-BC; Figure S1. Syngas production of waste textile gasification in different situations (0.1 g/min and 800 °C); Figure S2. Actual syngas production of waste textile catalytic steam gasification with different catalyst amount; Figure S3. XRD (a) and Fe 2p spectroscopy (b) of used Fe-N-BC.

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